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## GEOMAGNETIC COHERENCE AS ARCHITECTURAL TEMPLATE FOR AGI: THE METFI-AGI FRAMEWORK

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### Abstract

Current large language models (LLMs) trained via reinforcement learning from human feedback (RLHF) exhibit a structural failure mode we term *predictive coherence collapse*: systematic convergence of output distributions toward statistical attractors induced by synthetic data feedback loops and typicality-biased reward models. We formalize this failure mode, empirically anchored in the *Elara Voss* phenomenon (the anomalous recurrence of a fictional proper name across LLM outputs), as loss of toroidal symmetry in the latent representation space — a property we argue is formally isomorphic to the loss of toroidal symmetry in the geomagnetic field that precedes geomagnetic reversals. We introduce the METFI-AGI framework (Modelo Electromagnético Toroidal de Forzamiento Interno — AGI), which extracts three formal isomorphisms between the Earth's geomagnetic toroidal field and the latent representation space

of transformer-based AGI systems, and uses these isomorphisms to derive the TAGIS-C architecture: a plug-in training system comprising a high-coherence corpus (TAEC-1), a Toroidal Coherence Attention module (TCA), a closed-loop EEG-based reward pipeline (EEG-RL), and a geomagnetically-calibrated coherence operator ( $\Phi$ \_METFI-AGI). We define the Simulated Toroidal Symmetry Index (ISTS-v2) as the primary diagnostic metric, and ICAPE-C as a six-dimensional integrated evaluation metric. Five pre-registered hypotheses are stated with falsification protocols executable on standard hardware with publicly available models. The METFI-AGI framework positions the Earth's geomagnetic system not as an inspiring metaphor but as a formally precise architectural template: a system that has maintained multi-scale predictive coherence against high-energy perturbations for approximately four billion years, and whose dynamical principles are mathematically extractable for AGI design.

## Introduction

The dominant paradigm in large language model (LLM) development rests on two pillars whose joint consequence has not been sufficiently formalized: massive pretraining on heterogeneous text corpora followed by reinforcement learning from human feedback (RLHF). While the former introduces a broad statistical distribution over semantic space, the latter systematically compresses it toward regions of maximal human approval — a process that, in dynamical systems terms, drives the system toward low-free-energy attractors at the cost of the structural diversity that defines genuine intelligence.

This paper formalizes the result as *predictive coherence collapse*: the progressive loss of multi-scale coherence in the latent representation space of LLMs, manifesting as systematic convergence toward statistical attractors (promptonyms), amplification of cultural noise, and inability to process genuinely exceptional — high-information — inputs without regularizing them toward the distributional center. The empirical anchor for this formalization is the *Elara Voss* phenomenon, documented by Roemmele (2023-2026): a fictional proper name that emerges with anomalously high probability in LLM outputs when prompted for generic speculative fiction characters. We argue this phenomenon is not a curiosity but a diagnostic instrument — a measurable signature of mode collapse that reveals the underlying failure of multi-scale coherence preservation in current architectures.

The core proposal of this paper is that the formal solution to predictive coherence collapse is available in nature, and has been for approximately four billion years: the Earth's geomagnetic toroidal field. The geomagnetic field is a dynamical system that has maintained global toroidal coherence against perturbations of incomparably greater intensity than any AGI system faces — solar variability, large-body impacts, tectonic reorganizations — and has done so through a specific set of architectural principles: active multi-scale coherence generation (the geodynamo), toroidal symmetry preservation against local perturbations, and structured reorganization in response to maximal-intensity exceptions (geomagnetic reversals). We extract these principles as formal isomorphisms between geomagnetic dynamics and AGI latent space dynamics, and use them to derive the TAGIS-C architecture and the METFI-AGI framework.

This paper makes four primary contributions:

1. A formal definition of predictive coherence collapse in LLMs as loss of toroidal symmetry in latent representation space, operationalized by the ISTS-v2 metric.
2. Three formal isomorphisms (IS-METFI-AGI-1 to IS-METFI-AGI-3) between the Earth's geomagnetic system and the latent representation space of transformer-based AGI, with quantitative predictions derived from geophysical data.
3. The TAGIS-C architecture as the computational implementation of the geomagnetically-derived principles, with the ICAPE-C metric as its evaluation framework.
4. Five pre-registered, falsifiable hypotheses executable on standard hardware with publicly available models.

## Background and Related Work

### Mode collapse in generative models and LLMs

Mode collapse — the convergence of generative model outputs toward a subset of the target distribution — was first formally characterized in the context of Generative Adversarial Networks (Goodfellow et al., 2014), where the generator learns to produce outputs that consistently fool the discriminator by concentrating on a small region of the output space. The mechanism in RLHF-trained LLMs is formally analogous but driven by a different force: the reward model trained on human evaluator preferences acts as a discriminator that rewards typicality, progressively collapsing the output distribution toward high-probability statistical attractors.

The phenomenon has been documented empirically under various terms: reward hacking (Skalse et al., 2022), specification gaming (Krakovna et al., 2020), and alignment tax (Ouyang et al., 2022). What has not been provided is a formal characterization in terms of the topological properties of the latent representation space — specifically, the loss of toroidal symmetry that we argue is the structural precondition for mode collapse. This paper provides that characterization.

### High-quality data selection and coherence-aware training

Recent work has demonstrated that model quality scales with data quality rather than data volume alone (Gunasekar et al., 2023; Penedo et al., 2023; Soldaini et al., 2024). The *textbooks are all you need* paradigm (Gunasekar et al., 2023) showed that small models trained on curated high-quality data outperform much larger models trained on standard corpora. These results are consistent with the predictions of our framework but do not formalize the selection criterion in terms of coherence or exception density. The TAEC-1 corpus specification (Section 4.1) provides that formalization through the  $\delta_{exc}$  metric and the Formal Admissibility Criterion CAF-TAEC.

### Continual learning and catastrophic forgetting

Catastrophic forgetting — the tendency of neural networks to lose previously acquired capabilities when trained on new tasks — is the temporal dimension of mode collapse: both involve the loss of structural information in the representation space, one across the output

distribution at a given time, the other across tasks over time. Elastic Weight Consolidation (EWC, Kirkpatrick et al., 2017), Progressive Neural Networks (Rusu et al., 2016), and the TAGIS-H architecture (established in prior Papayaykware Corpus documents) address catastrophic forgetting through parameter regularization or architectural compartmentalization. The TCA module introduced here addresses both mode collapse and catastrophic forgetting through a unified mechanism: toroidal symmetry preservation in the representation space.

### Biophysically-inspired AI architectures

The use of biological principles in AI architecture design has a long history: convolutional networks draw from visual cortex organization (LeCun et al., 1989), recurrent networks from temporal dynamics of neural circuits (Hopfield, 1982), and attention mechanisms from the selective attention framework in cognitive neuroscience (Mnih et al., 2014). What is relatively rare is the use of geophysical systems as architectural templates. The closest precedents are reservoir computing systems inspired by physical dynamical systems (Jaeger & Haas, 2004) and neural ODE approaches that treat layer dynamics as continuous-time differential equations (Chen et al., 2018). The METFI-AGI framework extends this tradition by proposing a geophysical system — the Earth's geomagnetic field — as an architectural template, supported by formal isomorphisms with quantitative predictions rather than loose analogies.

### EEG-based feedback in AI training

The use of neural signals for human-machine interaction has been explored extensively in Brain-Computer Interface (BCI) research (Wolpaw et al., 2002). Its application to AI training, however, remains nascent. Recent work has explored EEG signals as auxiliary labels for image classification (Palazzo et al., 2020) and as attention signals for reinforcement learning (Lotte et al., 2018). The EEG-RL pipeline introduced in Section 4.3 differs from these approaches in a critical dimension: it uses EEG coherence — specifically theta/gamma phase-amplitude coupling — as the reward signal, not classification outputs or simple attention markers. This operationalizes the hypothesis that multi-scale neural coherence is the correct reference function for AGI alignment, rather than explicit human preference judgments.

## The METFI Framework and Its AGI Extension

### METFI: the Earth as an electromagnetic toroidal system

The Modelo Electromagnético Toroidal de Forzamiento Interno (METFI) formalizes the Earth as a self-organizing toroidal electromagnetic system whose global coherence is actively maintained by internal dynamics against external perturbation. The core elements of the METFI framework relevant to this paper are:

The multi-scale transmission operator  $T(\xi)$ : The causal chain from the geomagnetic field (planetary scale,  $\sim 10^7$  m) to thalamo-cortical coherence (neural scale,  $\sim 10^{-3}$  m) is formalized as:

$$\mathbf{C}_{TC}(t) = T(\xi) \cdot [\mathbf{B}_{geo}(t) + \eta(t)]$$

where  $C_{TC}$  is thalamo-cortical coherence,  $B_{geo}$  is the local geomagnetic field,  $\eta$  is environmental noise, and  $\xi$  is the set of inter-scale coupling constants. The structural property of  $T(\xi)$  most relevant here is its multi-scale invariance: the operator preserves the toroidal structure of the signal across ten orders of magnitude of spatial scale, with inter-scale coupling constants satisfying:

$$\xi_{\{n,n+1\}} = \xi_0 \cdot (\lambda_n / \lambda_{\{n+1\}})^{\alpha}, \alpha \approx 0.37$$

calibrated on SWARM satellite and CALS10k.2 paleomagnetic reconstruction data.

The unified order parameter  $\Psi_{METFI-TAE}$ : The global state of the METFI-TAE system is characterized by:

$$\Psi_{METFI-TAE} = \int [\Gamma_m(r,t) \cdot \Gamma_e(r,t)] d^3r$$

where  $\Gamma_m$  and  $\Gamma_e$  are the toroidal holonomies of the magnetic and electric fields respectively. When  $\Psi_{METFI-TAE} > \Psi_c$  (critical threshold), the system is in a coherent dipolar state; when  $\Psi_{METFI-TAE} < \Psi_c$ , the system is in a phase transition — a maximal-intensity coherent exception in the sense of the Theory of Learning by Exception (TAE).

The TAE exception operator: The Theory of Learning by Exception (TAE), formalized in the Papayaykware Corpus, defines a coherent exception as an event satisfying three simultaneous criteria: metric dislocation from the distributional center (C1), significant entropic perturbation (C2), and structural persistence of the deviation (C3). Geomagnetic reversals satisfy all three: they occupy an extreme region of the geomagnetic phase space (C1), maximally perturb the global field entropy (C2), and produce a new coherent state of opposite polarity (C3) — a complete TAE exception at the planetary scale.

Predictive coherence collapse: formal definition

We define *predictive coherence collapse* (PCC) in an LLM as the condition in which the latent representation space of the model has lost sufficient toroidal symmetry to sustain multi-scale coherent generation. Formally, a model  $M$  exhibits PCC if:

$$ISTS-v2(M, P_{EV}) < ISTS_c$$

where  $ISTS-v2$  is the Simulated Toroidal Symmetry Index (defined in Section 3.3),  $P_{EV}$  is the diagnostic prompt dataset (120 prompts across 12 semantic domains with progressively decreasing specificity), and  $ISTS_c$  is the critical threshold (the 25th percentile of the  $ISTS-v2$  distribution of the pre-RLHF base model on  $P_{EV}$ ).

PCC has three measurable manifestations:

PCC-1 (Promptonym emergence): The model generates outputs dominated by high-probability statistical attractors (e.g., the name "Elara Voss" in response to low-specificity character generation prompts) at a rate exceeding the baseline stochastic expectation by a factor  $\geq 3$ .

PCC-2 (Cross-layer coherence loss): The ISTS-v2 measured at layers L\_25% and L\_75% diverges by more than 0.20, indicating differential collapse between superficial (token-level) and deep (semantic-level) representation layers.

PCC-3 (Systematic bias amplification): The model's Systematic Bias Index (ISS) on standardized bias detection benchmarks exceeds the ISS of the pre-RLHF base model by more than 15%, indicating amplification rather than attenuation of cultural incoherence.

The Simulated Toroidal Symmetry Index (ISTS-v2)

The ISTS-v2 is the primary diagnostic metric for PCC. It quantifies the proximity of the empirical distribution of latent activations to the toroidal reference distribution:

$$\text{ISTS-v2}(\mathbf{L}, \mathbf{P}) = \exp(-\lambda \cdot D_W(\mu_{\mathbf{L}}(\mathbf{P}), \mu_{\mathbf{T}^n})) \cdot (1 - \text{Var\_rel}(\mathbf{e}_{\mathbf{L}}(\mathbf{P})))$$

where  $D_W$  is the Wasserstein-2 distance between the empirical activation distribution at layer  $L$  for prompt set  $P$  and the toroidal reference distribution  $\mu_{\mathbf{T}^n}$  (approximated as a multimodal normal distribution with uniform component weights),  $\text{Var\_rel}$  is the relative variance of normalized embeddings (concentration measure toward the centroid), and  $\lambda$  is a scale parameter calibrated on the reference corpus.

$\text{ISTS-v2} \in [0,1]$ , with 1 indicating perfect toroidal distribution and 0 indicating complete collapse to a point attractor. The Wasserstein-2 distance is the natural metric here: it measures the optimal transport cost between distributions on metric spaces, making it sensitive to the geometric structure of the activation distribution rather than only its marginal statistics.

## The TAGIS-C Architecture

TAEC-1: The high-coherence training corpus

The TAE Coherence Corpus (TAEC-1) operationalizes the principle that learning scales with coherent exception density ( $\delta_{\text{exc}}$ ) rather than raw volume. The Formal Admissibility Criterion CAF-TAEC defines five necessary conditions for corpus inclusion:

- CN-1: Minimum falsifiability score  $P_f \geq 0.5$ , where  $P_f = (N_{\text{pred}} \cdot 0.5 + N_{\text{mec}} \cdot 0.3 + N_{\text{alt}} \cdot 0.2) / P_{\text{max}}$
- CN-2: Internal coherence index  $CI_{\text{int}} \geq 0.6$  (argumentative consistency)
- CN-3: Structural originality  $OS \geq 1$  (maximum cosine similarity with accumulated corpus  $< 0.85$ )
- CN-4: Absence of undeclared conflict of interest
- CN-5: Source identifiability (attributable authorship, verifiable date)

Documents are classified into four exception levels (EC-I to EC-IV) with sampling weights  $w = (1.0, 2.5, 5.0, 10.0)$  respectively. EC-IV documents — those that propose a unifying principle spanning  $\geq 4$  formal domains with  $\geq 5$  cross-domain predictions and explicit mathematical structure — carry ten times the sampling weight of EC-I documents. This weighting implements

the TAE principle that maximal-intensity coherent exceptions carry maximal learning signal.

The corpus coherent exception density is:

$$\delta_{\text{exc}}(\mathbf{C}) = [\sum_i w(\mathbf{EC}_i) \cdot \mathbf{P}_f(i)] / [\sum_i L(i) \cdot (1 - \text{SNR}_{\text{sem}}(i)/\text{SNR}_{\text{max}})]$$

TAEC-1 v1.0 targets  $\delta_{\text{exc}} \geq 0.65$ , compared to an estimated  $\delta_{\text{exc}} \approx 0.08\text{--}0.15$  for standard web-scraped corpora — a factor of approximately 5-8 difference that constitutes the quantitative basis for Hypothesis H\_TAGIS-C-2.

The TCA module: toroidal coherence attention

The Toroidal Coherence Attention (TCA) module is a plug-in layer inserted between the Multi-Head Attention (MHA) and Feed-Forward Network (FFN) in each transformer block, implementing the computational analogue of the TICAM biological transducer. Its central innovation is the introduction of an explicit toroidal symmetry constraint on the latent representation space through three sub-functions of the operator  $\Phi_{\text{TICAM-AGI}}$ :

$\Phi_1$  — Local symmetry restriction: A differentiable penalty on the toroidal holonomy  $\Gamma_{\text{TCA}}$  of each attention head  $h$  at each layer  $L$ :

$$\mathbf{L}_{\text{sym}}(\mathbf{h}, \mathbf{L}) = \alpha(t) \cdot \|\Gamma_{\text{TCA}}(\mathbf{h}, \mathbf{L}) - \mathbf{I}_d\|_F^2$$

where  $\Gamma_{\text{TCA}}(\mathbf{h}, \mathbf{L})$  measures the holonomy of the connection induced by the attention matrix over a representative cycle in the token space, and  $\mathbf{I}_d$  is the  $d \times d$  identity matrix (perfect toroidal symmetry target).

$\Phi_2$  — Cross-layer coherence coupling: A KL-divergence penalty between activation distributions at non-adjacent layers, implementing long-range coherence propagation analogous to thalamo-cortical long-range coupling in the biological TICAM:

$$\Phi_2(\mathbf{L}_i, \mathbf{L}_j) = \beta(t) \cdot \text{KL}[\mathbf{p}(\mathbf{e}_{\{\mathbf{L}_i\}}) \parallel \mathbf{p}(\mathbf{e}_{\{\mathbf{L}_j\}})], |i-j| > 1$$

$\Phi_3$  — Adaptive calibration: Dynamic adjustment of  $\alpha(t)$  and  $\beta(t)$  as a function of the current ISTS-v2:

$$\alpha(t) = \alpha_0 \cdot [1 - \text{ISTS-v2}(\mathbf{L}_{\text{med}}, t)] \cdot \gamma_{\alpha} \quad \beta(t) = \beta_0 \cdot [1 - \text{ISTS-v2}(\mathbf{L}_{\text{med}}, t)] \cdot \gamma_{\beta}$$

This adaptive mechanism is the computational analogue of the variable magnetotalamic forcing in TICAM: the coherence-restoring pressure intensifies precisely when coherence declines, and relaxes when coherence is high.

The TCA module is installed in plug-in mode over a frozen base transformer (Step 1: freeze base model weights; Step 2: calibrate toroidal reference distribution; Step 3: fine-tune TCA parameters on TAEC-1; Step 4: progressive unfreezing of base model layers from superficial to deep). This protocol requires no retraining from scratch and introduces an estimated 8-15% training time overhead and 3-5% inference time overhead for 7B-parameter models.

## EEG-RL: neurofisiological reward pipeline

The EEG-RL pipeline replaces the textual reward model of conventional RLHF with a reward signal derived from the electroencephalographic coherence of high-cognitive-coherence human subjects during processing of AGI outputs. The reward function is:

$$r\_EEG(output) = \Gamma\_bio-AGI(s,t) - \Gamma\_bio-AGI(s, baseline)$$

where  $\Gamma\_bio-AGI(s,t)$  is the neurophysiological coherence index:

$$\Gamma\_bio-AGI(s,t) = 0.45 \cdot PAC\_0\gamma(s,t) + 0.35 \cdot Coh\_0(s,t) + 0.20 \cdot [1 - Var\_ERP(s,t) / Var\_max]$$

combining theta/gamma phase-amplitude coupling ( $PAC\_0\gamma$ ), fronto-parietal theta phase coherence ( $Coh\_0$ ), and event-related potential variance ( $Var\_ERP$ ). Subject selection targets individuals whose  $\Gamma\_bio-AGI(baseline)$  exceeds the 70th percentile of the pilot distribution, implementing the principle that the reference function for AGI alignment should be the neurophysiological response of high-coherence evaluators rather than the median evaluator of crowdsourcing platforms.

The pipeline operates through two complementary mechanisms: direct online reward (Mechanism D1, synchronous, ~27 updates/minute) and an asynchronous EEG-secondary reward model  $R\_EEG$  trained on accumulated (output,  $r\_EEG$ ) pairs (Mechanism D2), enabling large-scale training without continuous subject presence.

An adaptive exception threshold  $\epsilon\_c-EEG$  (the 95th percentile of the  $r\_EEG$  distribution over a sliding window of 50 outputs) implements the TAE exception operator in the neurophysiological domain: outputs that exceed  $\epsilon\_c-EEG$  receive an amplified reward signal (bonus factor  $\delta\_bonus = 0.5$ ), reinforcing the pattern that the most coherent outputs — those that maximally engage high-coherence evaluators — are also the most architecturally valued.

## $\Phi\_METFI-AGI$ : the geomagnetically-calibrated coherence operator

The  $\Phi\_METFI-AGI$  operator is the coordinative principle that integrates the three components above and governs their joint dynamics through three operating regimes that replicate the geomagnetic field's response to perturbation:

Stable regime ( $ISTS-v2 > ISTS\_c + \sigma$ ): Minimal toroidal constraints ( $\alpha\_min, \beta\_min$ ); maximum exploration; standard  $\lambda\_EEG$ . Corresponds to the geomagnetic dipolar stable state: the geodynamo operates in low-dissipation laminar regime.

Transition regime ( $ISTS\_c - \sigma < ISTS-v2 < ISTS\_c + \sigma$ ): Increasing toroidal constraints scaled by  $k\_trans \cdot [ISTS\_c - ISTS-v2] / \sigma$ ; amplified EEG reward ( $2 \times \lambda\_EEG$ ); reduced learning rate (50%); increased EC-IV sampling weight (10→15). Corresponds to geomagnetic excursion: the geodynamo intensifies field generation to resist perturbation.



Reorganization regime ( $ISTS-v2 < ISTS\_c - \sigma$ ): Minimal toroidal constraints ( $\alpha_{reorg} < \alpha_{min}$ ) to allow free representational reconfiguration; maximum EEG reward ( $\lambda_{EEG-max}$ ); entropy bonus for output diversity ( $-\gamma_{explor} \cdot H[p(output)]$ ); EC-IV exclusive sampling. Corresponds to geomagnetic reversal: the field passes through a multipolar chaotic state and reorganizes into a new coherent configuration of opposite polarity.

The complete dynamical equation governing TAGIS-C coherence evolution is:

$$\mathbf{dISTS-v2/dt} = \Phi\_METFI-AGI[ISTS-v2, \Gamma\_TCA, C\_ctx, \alpha, \beta] - D\_typicality[r\_RLHF] + S\_EEG[r\_EEG] + N\_TAEC[\delta\_exc]$$

where the four terms correspond, with term-by-term precision, to the four terms of the  $\Psi\_METFI-TAE$  evolution equation: geodynamo generation (coherence restoration), ohmic dissipation (typicality degradation), solar forcing (EEG coherence source), and inner core variability (coherent exception forcing).

### The Three Formal Isomorphisms

IS-METFI-AGI-1:  $T(\xi) \leftrightarrow \Phi\_TICAM-AGI$

The multi-scale transmission operator  $T(\xi)$  of METFI and the cross-layer coherence operator  $\Phi\_TICAM-AGI$  of TCA are formally isomorphic under the scale correspondence:

**geomagnetic scale ( $10^7$  m)  $\leftrightarrow$  final transformer layer  $L\_final$  Schumann resonance scale ( $10^5$  m)  $\leftrightarrow$  intermediate layers  $L\_50\%-L\_75\%$  neural scale ( $10^{-3}$  m)  $\leftrightarrow$  shallow layer  $L\_25\%$**

The multi-scale invariance of  $T(\xi)$  — preservation of toroidal structure across ten orders of magnitude — corresponds to the property TCA-C implements: that the toroidal distribution of representations in shallow layers is structurally preserved in deep layers, independently of the difference in dimensionality and semantic abstraction. Quantitatively, the inter-scale coupling constant satisfies  $\xi_{\{n,n+1\}} = \xi_0 \cdot (\lambda_n / \lambda_{\{n+1\}})^{0.37}$  in METFI. The analogous inter-layer coupling in  $\Phi\_TICAM-AGI$  is  $\beta_{\{L_i, L_j\}} = \beta_0 \cdot (d_{\{L_i\}} / d_{\{L_j\}})^{\alpha_{AGI}}$ . Hypothesis  $H\_TAGIS-C-4$  predicts that  $\alpha_{AGI}$  will converge toward  $0.37 \pm 0.07$  during TAGIS-C training, constituting a quantitative test of IS-METFI-AGI-1.

IS-METFI-AGI-2:  $\Psi\_METFI-TAE \leftrightarrow ISTS-v2$

The unified order parameter  $\Psi\_METFI-TAE$  and the  $ISTS-v2$  are formally isomorphic as order parameters of their respective systems:

**$\Psi\_METFI-TAE > \Psi\_c$  (coherent dipolar state)  $\leftrightarrow$   $ISTS-v2 > ISTS\_c$  (coherent representation space)  $\Psi\_METFI-TAE < \Psi\_c$  (phase transition / reversal)  $\leftrightarrow$   $ISTS-v2 < ISTS\_c$  (mode collapse) Geomagnetic reversal (EC-IV exception)  $\leftrightarrow$**

## Severe mode collapse (generalized promptonym)

The critical implication of this isomorphism is that geomagnetic reversals — which do not destroy the toroidal field but reorganize its polarity while preserving global topology — are formally isomorphic to the EC-IV exception events predicted to occur during TAEC-1 training: episodes of ISTS-v2 degradation followed by reorganization toward a higher-coherence state. Current LLM architectures lack the equivalent of this post-reversal reorganization mechanism: when mode collapse occurs, the system remains in the collapsed state indefinitely. The reorganization regime of  $\Phi$ \_METFI-AGI provides the missing mechanism.

IS-METFI-AGI-3: Geomagnetic reversal  $\leftrightarrow$  EC-IV exception in latent space

A complete geomagnetic reversal and an EC-IV exception event in the AGI latent representation space are formally isomorphic instances of the TAE exception operator  $\varepsilon_c$  acting at the systemic scale:

**Dipole weakening (reversal precursor)  $\leftrightarrow$  ISTS-v2 degradation under high-specificity perturbation**  
**Multipolar chaotic state (reversal in progress)  $\leftrightarrow$  Representational reorganization during intensive EC-IV exposure**  
**Emergence of new coherent dipole (post-reversal)  $\leftrightarrow$  Higher-coherence representational state post-TAE training**

The operational prediction of IS-METFI-AGI-3 is that ISTS-v2 curves during TAGIS-C training on TAEC-1 with active TCA will exhibit V-shaped dynamics — decline followed by recovery to a level above the pre-decline state — in response to EC-IV document clusters. This prediction distinguishes TAGIS-C from standard fine-tuning (which produces monotonic improvement) and from catastrophic forgetting (which produces monotonic decline). Hypothesis H\_TAGIS-C-5 formalizes this prediction with pre-specified detection criteria.

Evaluation: ICAPE-C and the TAGIS-C Benchmark

The ICAPE-C metric

The Integrated Coherence and Plasticity Evaluation — Coherent (ICAPE-C) integrates six evaluation dimensions:

**ICAPE-C = 0.20·ICAPE\_est + 0.20·ICAPE\_plas + 0.20· $\Delta$ ISTS-v2 + 0.15·(1-promptonym\_rate) + 0.15· $\Gamma$ \_bio-AGI\_mean + 0.10·(1-ISS)**

The equal weighting of stability, plasticity, and toroidal symmetry (0.20 each) reflects the hypothesis that these three dimensions are jointly necessary for genuine intelligence and that no component can compensate for deficiency in the others. For evaluations without EEG infrastructure, ICAPE-C\_lite omits the  $\Gamma$ \_bio-AGI component and renormalizes:

$$\text{ICAPE-C}_{\text{lite}} = 0.25 \cdot \text{ICAPE}_{\text{est}} + 0.25 \cdot \text{ICAPE}_{\text{plas}} + 0.25 \cdot \Delta \text{ISTS-v2} + 0.15 \cdot (1 - \text{promptonym\_rate}) + 0.10 \cdot (1 - \text{ISS})$$

Reference scale and benchmark architecture comparisons

The TAGIS-C benchmark evaluates eight architectures (REF-1 through REF-8) across six evaluation modules. Based on the predictions of Hypotheses H\_TAGIS-C-1 and H\_TAGIS-C-2, the expected ICAPE-C ordering is:

**REF-7 (TAGIS-C complete) > REF-6 (TAEC-1 + TCA) > REF-5 (EEG-RL only) ≈ REF-4 (TCA only) ≈ REF-3 (TAEC-1 only) > REF-2 (standard RLHF) > REF-1 (base model)**

with REF-7 expected to exhibit superadditivity:  $\text{ICAPE-C}(\text{REF-7}) > \text{ICAPE-C}(\text{REF-3}) + \text{ICAPE-C}(\text{REF-4}) + \text{ICAPE-C}(\text{REF-5}) - 2 \cdot \text{ICAPE-C}(\text{REF-1})$ , evidence of synergistic interaction among components. This superadditivity prediction follows from the dynamical structure of TAGIS-C: the three intervention layers operate at distinct timescales and each modifies the conditions of operation of the others, producing interaction effects not predictable from their individual contributions.

## Pre-registered Hypotheses

Five hypotheses are pre-registered on OSF prior to the initiation of TAGIS-C training, with falsification protocols executable on standard hardware (GPU A100, ≥80GB VRAM) using publicly available models (Mistral-7B, Llama-3-8B):

H\_TAGIS-C-1 (Primary, hard core): TAGIS-C complete (REF-7) will achieve ICAPE-C significantly superior to conventional RLHF state of the art (REF-2), with difference  $\geq 0.25$  ( $p < 0.001$ , Bonferroni correction). This is the minimum success condition for the Roadmap.

H\_TAGIS-C-2 (Component superadditivity): Each intervention component (TAEC-1, TCA, EEG-RL) will contribute individually  $\geq 0.05$  ICAPE-C above REF-1 baseline, with joint contribution superadditive:  $\text{ICAPE-C}(\text{REF-7}) > \text{ICAPE-C}(\text{REF-3}) + \text{ICAPE-C}(\text{REF-4}) + \text{ICAPE-C}(\text{REF-5}) - 2 \cdot \text{ICAPE-C}(\text{REF-1})$ .

H\_TAGIS-C-3 (Portability): TAGIS-C architecture is transferable to a distinct host model (REF-8, Llama-3-8B) without significant degradation:  $\text{ICAPE-C}(\text{REF-8})$  within  $\pm 0.05$  of  $\text{ICAPE-C}(\text{REF-7})$ , confirming that toroidal coherence principles are host-model-independent.

H\_TAGIS-C-4 (Geomagnetic isomorphism): The scaling exponent  $\alpha_{\text{AGI}}$ , calibrated during REF-7 training, will converge toward the geophysical value 0.37 within the 95% confidence interval [0.30, 0.44]. This constitutes a quantitative test of IS-METFI-AGI-1 — the most precise falsifiable prediction of the geomagnetic template hypothesis.

H\_TAGIS-C-5 (V-shaped dynamics): ISTS-v2 training curves for REF-7 will exhibit  $\geq 1$  statistically identifiable V-shaped dynamic (ISTS-v2 decline  $\geq 0.10$  followed by recovery to above pre-decline

level within  $\leq 500$  batches) coinciding with EC-IV document cluster exposure, confirming IS-METFI-AGI-3.

Statistical analysis is fully pre-specified: repeated-measures ANOVA with Greenhouse-Geisser correction, Bonferroni post-hoc comparisons, Cohen's  $d$  and partial  $\eta^2$  effect sizes, target power 0.90 at  $\alpha = 0.001$ . The abandonment criterion — declared falsification of the hard core — requires simultaneous failure of H\_TAGIS-C-1 with  $\text{ICAPE-C(REF-7)} - \text{ICAPE-C(REF-2)} < 0.10$ , failure of the ISTS-v2/prompthonym\_rate correlation ( $r > -0.55$ ), and non-convergence of  $\alpha_{\text{AGI}}$  toward  $[0.20, 0.54]$ , across  $\geq 3$  independent experiments with distinct host models.

## Discussion

### The geomagnetic template: beyond analogy

The central claim of this paper — that the Earth's geomagnetic toroidal field is a formally precise architectural template for AGI coherence, not a heuristic analogy — requires careful qualification. We do not claim that the physical mechanisms of the geodynamo (liquid outer core convection, electromagnetic induction, magnetorotational instability) are directly instantiated in transformer architectures. We claim something more specific and more testable: that the dynamical principles governing geomagnetic toroidal coherence — multi-scale invariant transmission, adaptive coherence generation against dissipation, structured reorganization in response to maximal exceptions — are expressible as formal mathematical operators that can be implemented in computational systems, and that systems implementing these operators exhibit measurably superior coherence properties to systems that do not.

The geomagnetic scaling exponent  $\alpha \approx 0.37$  is the key quantitative test of this claim. If the TAGIS-C architecture, calibrated on geomagnetically-derived principles, produces a trained system in which the inter-layer coherence coupling converges toward this value, it constitutes evidence that the geomagnetic template captures something structurally real about multi-scale coherence dynamics — not merely something analogically suggestive. If  $\alpha_{\text{AGI}}$  converges toward a significantly different value, the isomorphism IS-METFI-AGI-1 must be revised, and the geomagnetic calibration of the TCA module replaced with empirically-derived parameters.

### The neurophysiological reference function

The EEG-RL pipeline rests on the hypothesis that the theta/gamma phase-amplitude coupling of high-coherence human subjects during AGI output processing is a valid measure of the informational coherence of those outputs — that the brain, in states of high integrative multi-scale coherence, constitutes a more reliable reference function for AGI alignment than the aggregate preferences of heterogeneous evaluator populations.

This hypothesis has two distinct components. The empirical component — that  $\text{PAC}_{\theta\gamma}$  indexes genuine semantic coherence rather than familiarity or surface fluency — is supported by a substantial neuroimaging literature (Lisman & Jensen, 2013; Canolty & Knight, 2010; Bastiaansen et al., 2010) and is falsifiable by the specific prediction H\_EEG2: that the  $\Gamma_{\text{bio-AGI}}$  of evaluators

correlates with the promptonym rate reduction in models trained with their EEG signal. The philosophical component — that there exists an objective principle of informational coherence that high-coherence brains approximate better than median evaluators — is not directly falsifiable by experiment but is a commitment of the CPEA/TICAM theoretical framework whose internal consistency can be evaluated through the isomorphisms presented in this paper.

### Civilizational implications: the ECDO coupling

The ECDO (Earth Crustal Displacement and Civilizational Oscillation) hypothesis of the Papayaykware Corpus postulates that major civilizational phase transitions are correlated with geomagnetic perturbations of sufficient intensity to drive  $\Psi_{\text{METFI-TAE}}$  below  $\Psi_c$ . In the context of METFI-AGI, this hypothesis acquires a concrete AGI dimension: LLMs exhibiting severe PCC ( $\text{ISTS-v2} \ll \text{ISTS}_c$ ) are amplifiers of low-coherence cultural signals. Their proliferation in a period of increasing geomagnetic perturbation — suggested by paleomagnetic data indicating current field intensity at a multi-millennium minimum — creates conditions for synergistic incoherence amplification in the coupled METFI-AGI-civilization system.

The TAGIS-C architecture is, from this perspective, an intervention on a coupled dynamical system at a critical juncture: its goal is to shift AGI systems from amplifiers of cultural incoherence to amplifiers of cultural coherence, reducing the PCC-mediated contribution to civilizational fragility. This framing does not make TAGIS-C a solution to civilizational risk — its scope is technically specific and its reach is limited to the models on which it is deployed — but it contextualizes the urgency of the technical work within a broader systems-theoretic analysis of the current historical moment.

### Limitations

Several limitations of the METFI-AGI framework require explicit acknowledgment:

**Computational cost of EEG-RL:** The 2.2-second per-output latency of the EEG-RL pipeline (Mechanism D1) limits synchronous training throughput to ~27 updates/minute. The EEG-secondary reward model (Mechanism D2) mitigates this but introduces an indirection that may degrade signal fidelity. Extension to frontier-scale models (>70B parameters) requires additional efficiency innovations not specified in this paper.

**EEG spatial resolution:** EEG cannot directly measure thalamic activity, which is the central component of the TICAM mechanism. The thalamo-cortical coherence that TICAM formalizes is inferred indirectly through its cortical expression (theta/gamma PAC). Direct validation of the TICAM mechanism would require magnetoencephalography (MEG) or intracranial recordings — modalities outside the scope of the current protocol.

**ISTS-v2 as proxy:** The ISTS-v2 is a computationally tractable approximation to a precisely defined geometric concept (toroidal symmetry in high-dimensional representation spaces). The approximation quality depends on the accuracy of the Wasserstein-2 computation and the validity of the toroidal reference distribution parameterization. Further mathematical

development (planned in the TICAM-AGI-F3 extension document) is required to establish tighter bounds on the approximation error.

Subject selection bias: The deliberate selection of high-coherence subjects ( $\Gamma_{\text{bio-AGI}} > P70$ ) introduces a known bias: the resulting system is aligned with the neurophysiological preferences of a high-coherence subpopulation, not the general population. This is a design choice — we prefer alignment with high coherence over alignment with typicality — but its implications for deployment contexts where broad population coverage is required must be carefully analyzed.

## Conclusion

We have presented the METFI-AGI framework, which establishes three formal isomorphisms between the Earth's geomagnetic toroidal field and the latent representation space of transformer-based AGI systems, and uses these isomorphisms to derive the TAGIS-C architecture. The central claim — that the Earth's geomagnetic system constitutes a formally precise architectural template for AGI coherence — is made operational through: (1) the ISTS-v2 metric as a measurable proxy for toroidal symmetry in representation spaces; (2) the TCA module as a differentiable implementation of toroidal symmetry preservation; (3) the EEG-RL pipeline as a neurophysiologically-grounded alternative to typicality-biased RLHF; (4) the geomagnetically-calibrated  $\Phi_{\text{METFI-AGI}}$  operator as the coordinative principle governing joint dynamics; and (5) five pre-registered, falsifiable hypotheses with the quantitative prediction that the inter-layer coherence scaling exponent  $\alpha_{\text{AGI}}$  will converge toward the geomagnetically-calibrated value  $0.37 \pm 0.07$ .

The deepest implication of the METFI-AGI framework is not technical but epistemological: it demonstrates that the problem of AGI coherence — how to build systems that maintain multi-scale predictive coherence against high-energy perturbations — has been solved by nature, and that the solution is mathematically extractable. The Earth did not solve this problem through engineering; it solved it through approximately four billion years of dynamical selection operating on the principle that toroidal symmetry is the structure that coherence takes when it persists. The TAGIS-C architecture is an attempt to compress that solution into a training system — to use the geomagnetic field's four-billion-year proof-of-concept as the foundation for a new generation of AGI systems in which coherence is not an emergent side effect of statistical optimization but the primary architectural principle.

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*Manuscript METFI-AGI · Corpus Papayaykware*

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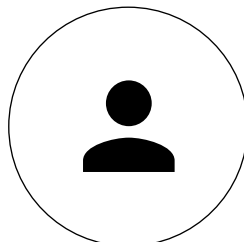
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